

BUILD WITH AI · CODE FOR COMMUNITIES — SECOND EDITION

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# CIVICPULSE BRICS

Scalable AI-Driven Infrastructure Prioritization for Digital Public Goods

*“From Citizen Voices to Actionable Infrastructure Intelligence.”*

TRACK: OPEN INNOVATION

Team: [Team Name] · Members: [Names + Roles]

CITIZENS -> AI -> SECTOR HEALTH MATRIX  
& TELEMETRY -> POLICYMAKERS

# Fragmented citizen voices delay action

## TODAY

- Voice, text, and messaging feedback is scattered across channels.
- Reports arrive in different languages and regional dialects.
- Unstructured reports are difficult to analyze and deduplicate at scale.
- Critical infrastructure hotspots stay hard to detect in real time.

1. Citizen Reports
2. Data Silos
3. Limited Visibility
4. Delayed Prioritization

### CORE GAP

**Citizen feedback exists.**

**Actionable intelligence doesn't.**

## THE SOLUTION

# From feedback to action

CivicPulse BRICS turns citizen reports into geospatially prioritized infrastructure intelligence.

### REPORT

Voice / Text / Custom Location & Auto-GPS

### UNDERSTAND

AI Transcription + Multilingual Translation

### CLUSTER

Geospatial Proximity Hotspots (10km DBSCAN)

### PRIORITIZE

Multi-Factor ML Priority Index (0–100 Scorer)

### ACT

AI Project Directives, Budgets & Policy Review

**Don't just collect complaints. Identify emerging infrastructure needs.**

## THE PLATFORM

# One platform. Five capabilities.

### 01 Citizen Portal

Voice + Text + Custom Location

Interactive GPS detection & BRICS country selector

### 02 Multilingual AI

Transcription + Translation

Automatic dialect synthesis & hazard classification

### 03 Geo-Intelligence

10km Proximity Clustering

DBSCAN grouping of isolated infrastructure requests

### 04 Priority Engine

Volume + Outage Duration + Sector Deficit + Disruption Intensity

Calculates unified 0–100 ML severity ranking across sectors

### 05 Policy Command Center

Sector Health Matrix + Autonomous Radar + AI Directives

Interactive radial dials, live threat sweep radar & policy review briefs

## TECHNICAL CORE

# The CivicPulse intelligence pipeline

01

VOICE /  
TEXT

02

GEMINI 2.5  
FLASH

03

0-100 ML  
SEVERITY

04

10KM  
DBSCAN

05

PRIORITY  
INDEX

06

AI POLICY  
BRIEF

Transcribe · Translate · Categorize · 0-100 Multi-Factor Severity Scorer · Geospatial Clustering

## TECHNICAL CORE

Gemini 2.5 Flash API + Scikit-Learn DBSCAN + Haversine Matrix + Zero-Setup SQLite / PostgreSQL



# From data to decision

**SECTOR MATRIX & RADAR**

Real-time radial gauges for Water, Roads, Power, Sanitation & autonomous threat radar sweep.

**LIVE FEED & TELEMETRY**

Anonymized, AI-translated citizen requests with live institutional status tickers.

**AI POLICY DIRECTIVES**

Autonomous project solutions with actionable budgets, timelines, and policy review citations.

1

50 Citizen Reports

2

1 Geographic Hotspot

3

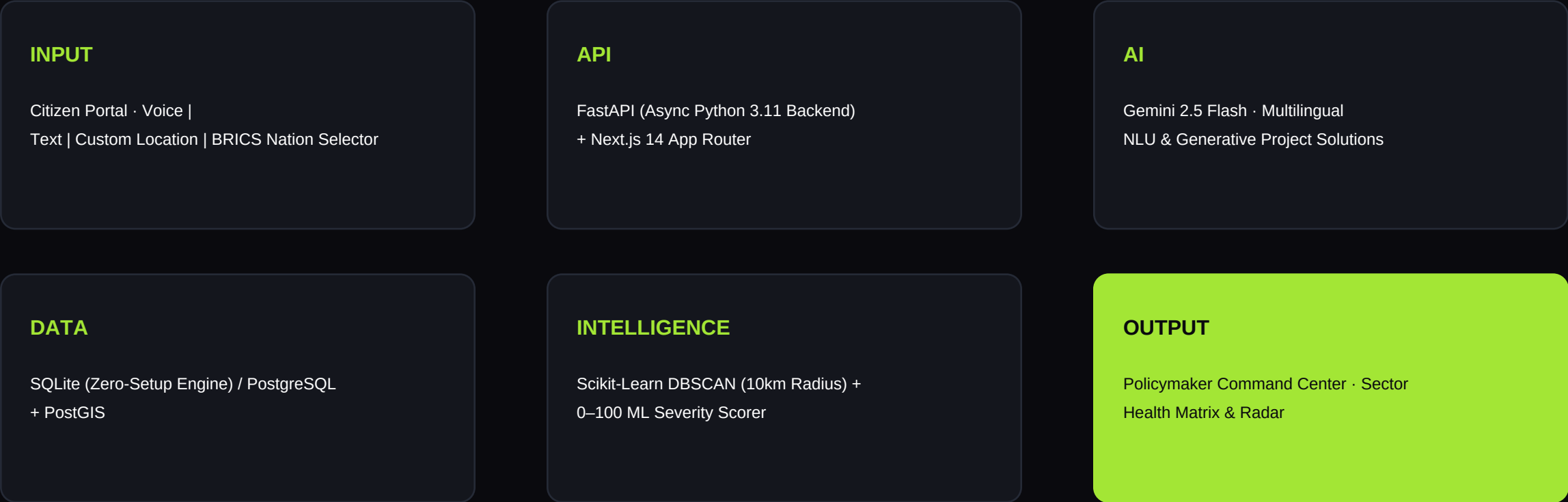
1 Priority Project

4

Policy Action

SYSTEM DESIGN

# Simple, scalable architecture



PRODUCTION-READY BY DESIGN

# Built for real-world civic systems

## PRIVACY

PII sanitization & anonymization  
before persistence

## SECURITY

JWT + HTTP-only cookies  
& input schema validation

## ACCESS CONTROL

RBAC: SuperAdmin | Policymaker  
| Analyst | Citizen

## SCALABILITY

Local ward -> Regional  
-> National -> BRICS-scale

## SUCCESS METRICS

Processing Time · Cluster Quality · Multi-Factor ML Accuracy · PII Removal Effectiveness



## THE END-TO-END PROMISE

# One report -> one actionable decision

1. Citizen Voice & Text Report
2. AI Multilingual Audio Synthesis
3. Automatic Translation & Hazard Tagging
4. PII Sanitization & Anonymization
5. Geospatial Clustering (10km DBSCAN)
6. 0–100 Multi-Factor ML Severity Scoring
7. AI Project Recommendation with Budget
8. Sector Health Matrix & Policy Action

## REAL-WORLD EXAMPLE

“Pipeline rupture reported in Ward 4”

-> Hotspot detected (10km DBSCAN)

-> ML Severity 88/100 (Critical Hazard)

-> Project Brief auto-drafted with \$85,000 budget & 4-day timeline

## CIVICPULSE BRICS

*“Turning Citizen Voices into Actionable Infrastructure Intelligence.”*

Build with AI: Code for Communities — Second Edition

Multilingual AI · Geospatial Intelligence · Digital Public Goods